Geometry Review: Unit 6 — Circles

Answer Key

Station 1: Parts of a Circle, Circumference, and Area

1. $C = 10\pi,\ A = 25\pi$
2. $r = 7,\ C = 14\pi,\ A = 49\pi$
3. $r = 9,\ A = 81\pi$
4. $r = 8,\ d = 16$
5. (a) radius; (b) diameter; (c) chord; (d) tangent (touches the circle at one point); (e) secant (crosses the circle twice)
6. $C = 20\pi \approx 62.8$ ft
7. $64\pi - 36\pi = 28\pi$ in^2
8. The area is multiplied by 4
9. $28\pi \approx 88$ in
10. $81\pi - 25\pi = 56\pi$

Station 2: Arc Length and Sector Area

1. $2\pi;\ 6\pi$
2. $5\pi;\ 25\pi$
3. $2\pi;\ 8\pi$
4. $6\pi;\ 27\pi$
5. $10\pi;\ 60\pi$
6. $120°$
7. $150°$
8. $2\pi;\ 5\pi$
9. 8π in^2
10. 12

Station 3: Central Angles, Inscribed Angles, and Arcs

1. $80°$
2. $70°$
3. $62°$
4. $90°$
5. $50°,\ 100°,\ 210°$
6. $85°$ and $100°$ *opposite angles are supplementary*
7. $41°$
8. $90°$ $\quad \frac{70 + 110}{2}$
9. $45°$ $\quad \frac{130 - 40}{2}$
10. $50°$

Station 4: Chords, Tangents, and Segment Lengths

1. 13 $\quad 5^2 + 12^2 = 13^2$
2. $x = 5;\ PA = 19$
3. $x = 8 \quad 4 \cdot 6 = 3x$
4. $x = 15$
5. $x = 12 \quad 4 \cdot 9 = 3x$
6. 9; 5 $\quad 6^2 = 4 \cdot w$
7. 24 $\quad$ *half-chord* $= \sqrt{13^2 - 5^2} = 12$
8. 6
9. $x = 9;\ AB = 23$
10. 8